M&AE 5070 - Flight Dynamics

Chapter 6 - Lateral Equations of Motion

Spring 2026

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6 Summary **17**

This chapter introduces the lateral equations of motion, with a discussion on the various lateral force and moment coefficients appearing in them. But before, we need to have a closer look at reference frames.

1 Reference frames and how they are related

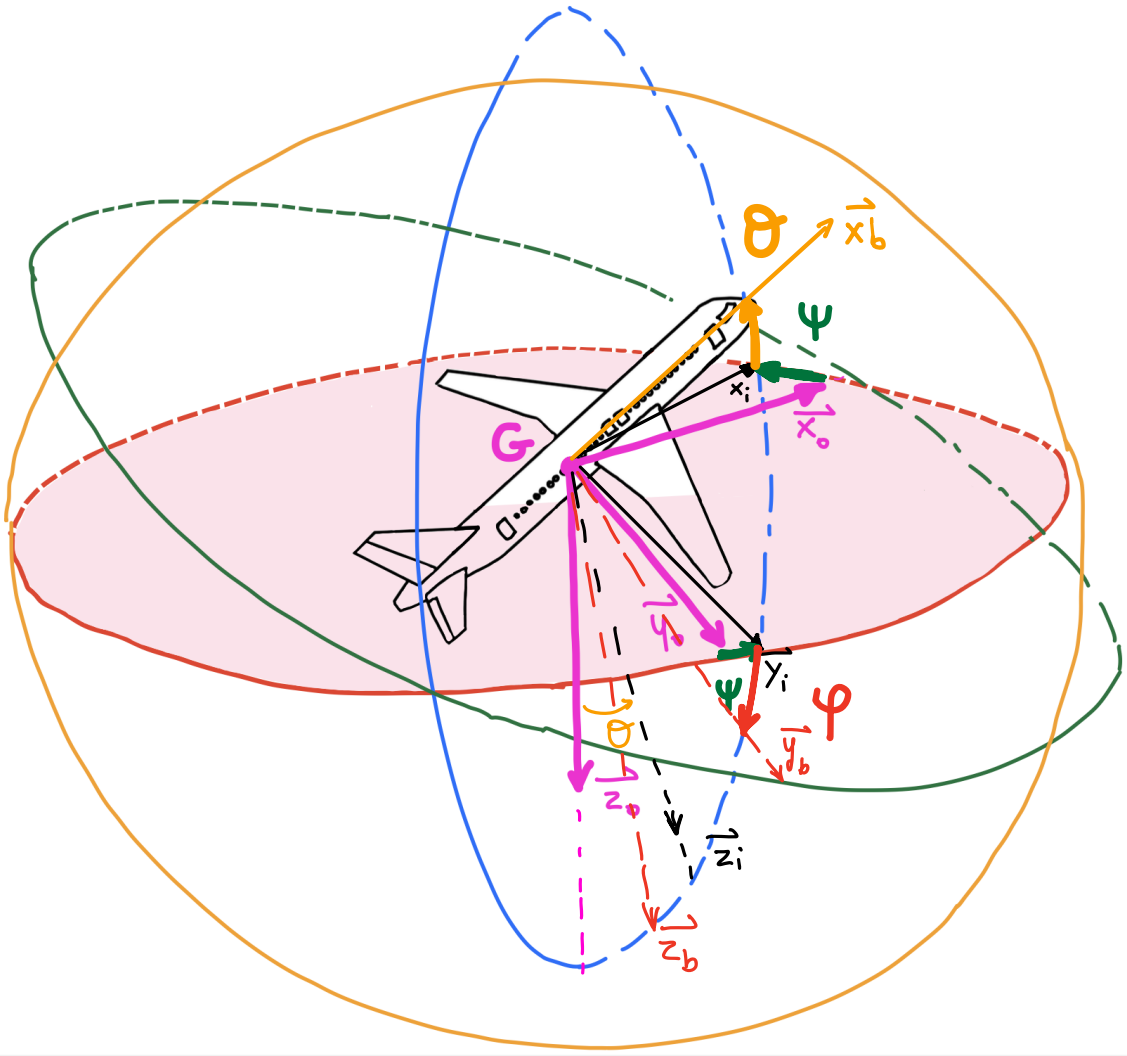
1.1 Reference frame definitions

- **Earth-based frame of reference** R_o - Origin O is a fixed (arbitrary) point on the ground
  - $\overrightarrow{z_o}$: in the direction of the gravitational force seen by the plane
  - $\overrightarrow{x_o}$: horizontal, pointing toward true or magnetic North
  - $\overrightarrow{y_o}$ completes the basis vectors according to the right hand rule.
- **Aircraft-based frame of reference** R_b - Origin G, center of gravity of the plane (even if not technically fixed due to fuel consumption)
  - $\overrightarrow{x_b}$: oriented forward, along the fuselage axis
  - $\overrightarrow{z_b}$: in the plane of symmetry of the aircraft, oriented toward the "floor" of the aircraft
  - $\overrightarrow{y_b}$: perpendicular to the plane of symmetry of the aircraft, oriented towards the pilot's right.
- **Aerodynamic frame of reference** R_a - Origin G, same as aircraft-based reference frame
  - $\overrightarrow{x_a}$: oriented in the direction of the aircraft velocity
  - $\overrightarrow{y_a}$ and $\overrightarrow{z_a}$: obtained by rotation of the aircraft-based reference frame by 2 angles, α_a (aircraft angle of attack) and β_a (side slip angle)

1.2 Switching from R_o to R_b

Switching reference frames from R_o to R_b requires 3 angles, corresponding to 3 successive rotations:

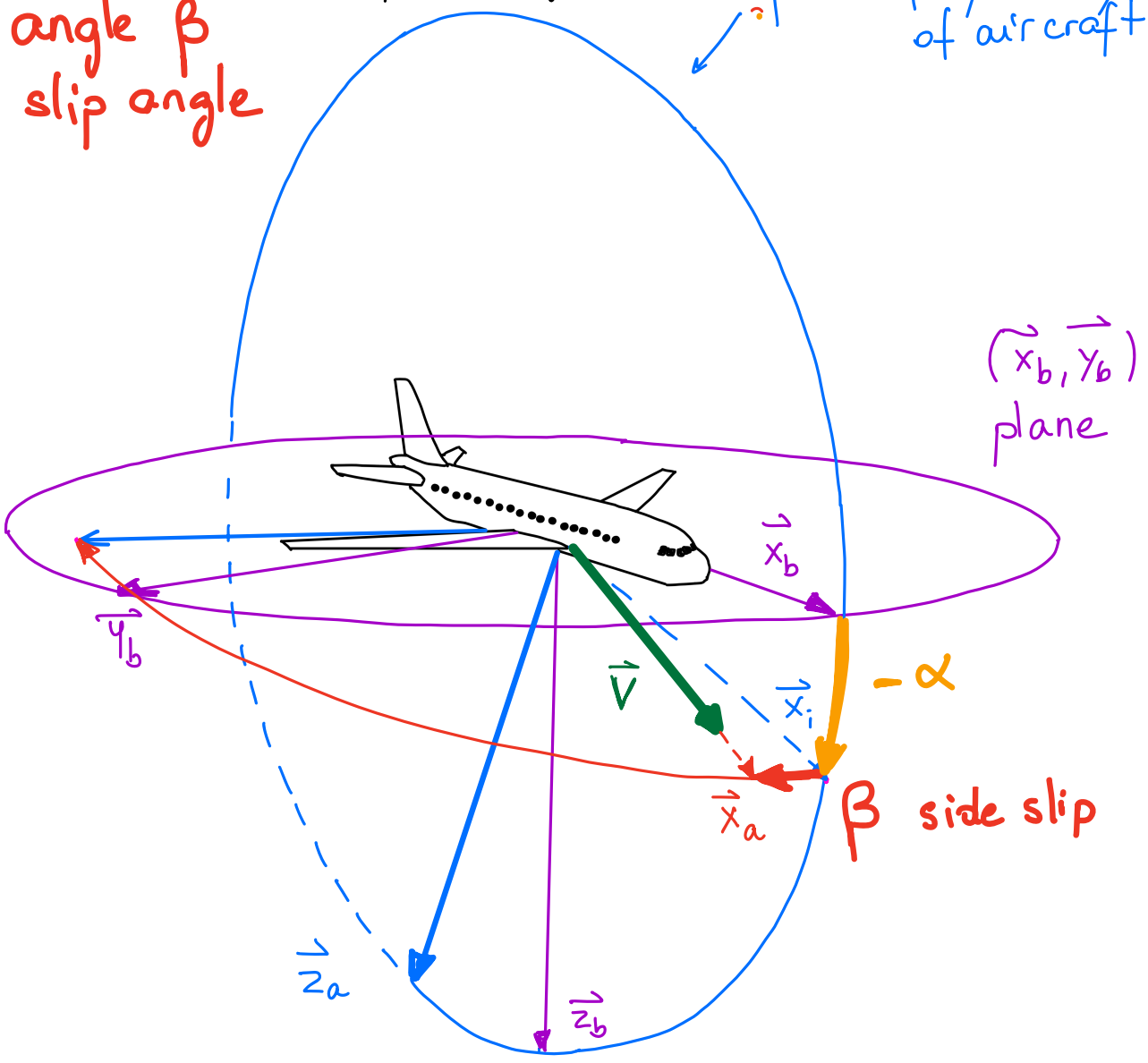
- 1st rotation: around $\vec{z}_o$, to bring $\vec{x}_o$ onto plane of symmetry of aircraft
  ANGLE Ψ : Heading / bearing angle
- 2nd rotation: around $\vec{y}_i$ (resulting from $\vec{y}_o$ through 1st rotation)
  so that $\vec{x}_i$ (from $\vec{x}_o$ through 1st rot) becomes $\vec{x}_b$
  ANGLE θ : PITCH / ATTITUDE
- 3rd rotation: around $\vec{x}_b$ to bring $\vec{y}_i$ onto $\vec{y}_b$
  ANGLE φ : BANK ANGLE / ROLL ATTITUDE



1.3 Switching from R_b to R_a

Switching reference frames from R_b to R_a corresponds to bringing $\vec{x_b}$ onto $\vec{x_a}$: only need 2 rotations to do that:

- 1st rotation: around $\vec{y}_b$, to bring $\vec{z}_b$ to $\vec{z}_a$ angle $-\alpha$
- 2nd rotation: around $\vec{z}_a$, to bring $\vec{x}_i$ to $\vec{x}_a$



Conventions:

* $\alpha > 0$ if $\vec{x}_b$ is ABOVE the $(\vec{x}_a, \vec{y}_a)$ plane (pilot above the velocity vector)
* $\beta > 0$ if air blows on right side of fuselage (pilot feels wind on right cheek)

1.4 Forces and moment coefficients

In the aerodynamic frame of reference R_a, the force and moment coefficients are written:

$$\overrightarrow{C_F}^a = \begin{pmatrix} C_x^a \\ C_y^a \\ C_z^a \end{pmatrix} = \begin{pmatrix} -C_x \\ C_y \\ -C_z \end{pmatrix}$$

$-C_x$ → drag coefficient; $-C_z$ → lift coefficient

$$\overrightarrow{C_M}^a = \begin{pmatrix} C_\ell^a \\ C_m^a \\ C_n^a \end{pmatrix}$$

not using it

In the aircraft frame of reference R_b, the force and moment coefficients are written:

$$\overrightarrow{C_F}^b = \begin{pmatrix} C_x^b \\ C_y^b \\ C_z^b \end{pmatrix} = \begin{pmatrix} -C_A \\ C_Y \\ -C_N \end{pmatrix}$$

$-C_A$ → Axial force coefficient; C_Y → NOTE $Y \neq y$; $-C_N$ → Normal force coefficient

$$\overrightarrow{C_M}^b = \begin{pmatrix} C_\ell^b \\ C_m^b \\ C_n^b \end{pmatrix} = \begin{pmatrix} C_\ell \\ C_m \\ C_n \end{pmatrix}$$

C_m → moment coefficient in longitudinal equations

- For *longitudinal* dynamics, R_a typically used
- In contrast, for *lateral* dynamics, R_b typically used.

2 Decoupling lateral from longitudinal motion

To simplify the analysis of lateral aircraft dynamics, the lateral equations must be decoupled as much as possible from the longitudinal ones. Recall that the decoupling conditions for the longitudinal motion were rather simple: if all external forces acting on an aircraft are contained in the symmetry plane of that aircraft, all the lateral parameters involved in the general equations of motion just disappear:

- side-slip angle β
- yaw and roll velocities (ANGULAR velocities, rad/s): r (yaw), p (roll)
- bank/roll angle ϕ (or φ)

Unfortunately, the same is **not** true for the lateral equations: regardless of the level of simplification we adopt, some longitudinal parameters will always appear in the lateral equations of motion:

Velocity V, angle of attack α, pitch velocity q

To keep things as simple as possible, we will keep all longitudinal parameters constant (controlled by the pilot), and we will look at the lateral motion from this initial longitudinal state.

$V = \text{constant} = V_e$
$\alpha = \text{constant} = \alpha_e$
$q = \text{constant} = q_e = 0$
+ small side slip angle : $\cos\beta \approx 1$, $\sin\beta \approx \beta$

3 Matrix of inertia

$$\mathbb{I} = \begin{pmatrix} A & 0 & -E \\ 0 & B & 0 \\ -E & 0 & C \end{pmatrix}$$

M equation

* 0 are because aircraft is symmetric
* thickness in z $\ll$ other direction

$C \approx A + B$

$A = \int (y^2 + z^2)\, dm$
$B = \int (z^2 + x^2)\, dm$
$C = \int (x^2 + y^2)\, dm$

moments of inertia

$E = \int xz\, dm$ product of inertia

4 Lateral equations in the frame of reference of the aircraft R_b

The set of lateral equations consists of 2 moment equations for roll and yaw (net torque = rate of change of angular momentum), the lateral force balance, and a kinematic equation for the roll attitude ϕ.

- **R** - Equation for the roll moment, around $\vec{x_b}$:

$$A\dot{p} - (B - C)q_e r - E(\dot{r} + pq_e) = \frac{1}{2}\rho S\ell V_a^2 C_\ell \qquad (1)$$

p

- **Y** - Equation for the yaw moment, around $\vec{z_b}$:

$$C\dot{r} - (A - B)q_e p - E(\dot{p} + rq_e) = \frac{1}{2}\rho S\ell V_a^2 C_n \qquad (2)$$

r

lateral force

- **LF** - Balance of forces along $\vec{y_b}$:

$$mV\left(\dot{\beta} + r\cos\alpha_e - p\sin\alpha_e\right) = mg\sin\phi\cos\theta_e + \frac{1}{2}\rho S V_a^2 C_Y \qquad (3)$$

β

- **Kϕ** - Kinematic equation for the roll attitude ϕ around $\vec{x_b}$:

$$\dot{\phi} = p + \tan\theta_e(r\cos\phi + q_e\sin\phi) \qquad (4)$$

ϕ

There are technically two additional equations for the bearing ψ and the lateral position of the aircraft, but those are fully decoupled and do not play any role on the dynamics, and are therefore of no interest here.

Simplifications

$$\begin{cases} q_e = 0 & \text{no pitching motion when looking at lateral motion} \\ E = 0 & \\ \alpha_e, \theta_e & \text{are small} \end{cases}$$

Under those assumptions, the equations above can be greatly simplified:

- **R** ROLL $\longrightarrow$ p : angular momentum around x_b

$$A\dot{p} = \frac{1}{2}\rho S \ell V^2 C_\ell$$

- **Y** YAW $\longrightarrow$ r : angular momentum around z_b

$$C\dot{r} = \frac{1}{2}\rho S \ell V^2 C_n$$

- **LF** Lateral force balance $\longrightarrow$ β : along y_b

$$mV(\dot{\beta} + r) = \underbrace{\frac{1}{2}\rho S \ell V^2 C_y}_{\text{aerodynamic lateral force}} + \underbrace{mg \sin\phi}_{\text{projection of weight along } y_b}$$

- **Kϕ**

$$\dot{\phi} = p$$

5 Non-dimensional lateral force and moment coefficients

The equations above are written in terms of lateral aerodynamic)force and moments) coefficients, for which we need models. We will adopt the same linear approach as what was done for the longitudinal coefficients. In theory, the lateral coefficients can be impacted by:

- side-slip angle β $\longrightarrow$ lateral angle of attack
- local angles linked to roll & yaw velocity : p, r
- controls : δ_ℓ (from ailerons), δ_n (from rudder)

The lateral force and momentum coefficients are linked to the **aircraft** frame of reference: and are linearized in the following general format:

Lateral force coefficient
- $C_Y = C_{Y_\beta}\beta + C_{Y_p}\left(p\frac{\ell}{V}\right) + C_{Y_r}\left(r\frac{\ell}{V}\right) + C_{Y_{\delta_\ell}}\delta_\ell + C_{Y_{\delta_n}}\delta_n$

roll moment coefficient
- $C_\ell = C_{\ell_\beta}\beta + C_{\ell_p}\left(p\frac{\ell}{V}\right) + C_{\ell_r}\left(\frac{r\ell}{V}\right) + C_{\ell_{\delta_e}}\delta_e + C_{\ell_{\delta_n}}\delta_n$

yaw moment coefficient
- $C_n = C_{n_\beta}\beta + C_{n_p}\left(\frac{p\ell}{V}\right) + C_{n_r}\left(\frac{r\ell}{V}\right) + C_{n_{\delta_e}}\delta_e + C_{n_{\delta_n}}\delta_n$

For example,

$C_{\ell_r}\left(\frac{r\ell}{V}\right) \equiv$ non-dimensional form of the ROLL moment created by the yaw angular velocity

The signs of the coefficients are very important to understand the dynamic behavior of the aircraft. They follow the same convention as the axes they are associated with:

C_Y (non-dimensional lateral force) > 0 if Y (lateral force) oriented in the direction of $\vec{y}_b$

C_ℓ (non-dim. roll moment) > 0 if $\frac{L}{\vec{x}_b}$ (roll moment) created around brings the RIGHT wing down

The rest of this chapter looks at each major contributions to C_Y, C_ℓ and C_n, and explores the underlying physics and expected signs of those contributions.

5.1 Lateral force coefficient C_Y

$\rightarrow$ C_{Y_r}, C_{Y_p} are neglected

The lateral force depends primarily on the side slip angle β, and the controls δ_ℓ and δ_n. There is no contribution from p and r.

$$C_Y = C_{Y_\beta}\beta + C_{Y_{\delta_\ell}}\delta_\ell + C_{Y_{\delta_n}}\delta_n$$

- Influence of side-slip: $C_{Y_\beta} = \dfrac{\partial C_Y}{\partial \beta}$

$\beta > 0 \Rightarrow$ velocity vector is to the right of symmetry plane (wind on "RIGHT Cheek" of pilot)

Sign of C_{Y_β}: aero force is oriented to the left (negative direction along $\vec{y}_b$)

$$\boxed{C_{Y_\beta} < 0}$$

The lateral force from side-slip is mostly due to the vertical stabilizer (VS)

(fuselage does contribute, but to lesser extent : neglect for now)

$\Rightarrow$ Analogy with main wings :

wing	lift force	angle of attack
$\Updownarrow$	$\Updownarrow$	$\Updownarrow$
VS	lateral force Y	side slip β

$$\text{lateral force } Y = f(\beta, AR_{VS}, \rho, V, S_{VS})$$

At the lifting-surface level, potential flow theory provides:

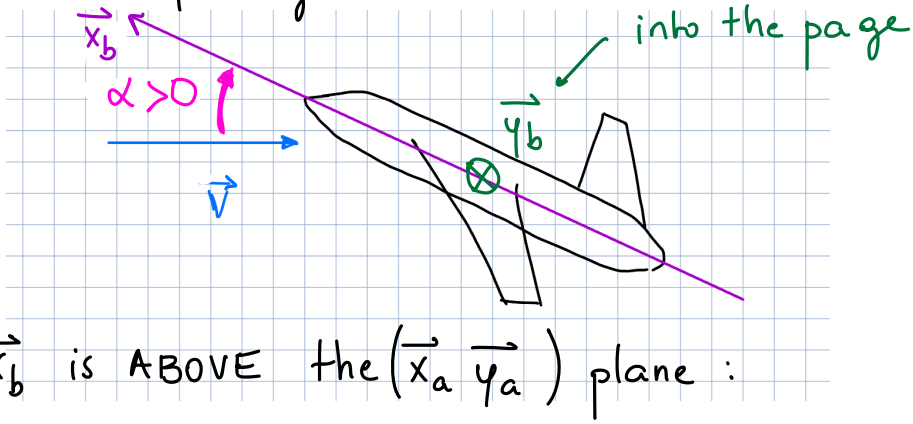
$C_{Z_\alpha} \Longleftrightarrow C_{Y_\beta}$

C_{Z_α}: $\dfrac{2\pi}{1+\frac{2}{AR_W}}$ $\qquad$ C_{Y_β}: $\dfrac{2\pi}{1+\frac{2}{AR_{VS}}}$

Notes on angles α and β

Angle of attack α : traditionally defined from airfoil configurations (in 2D) as the angle between the aerodynamic velocity vector and the chord of the airfoil.

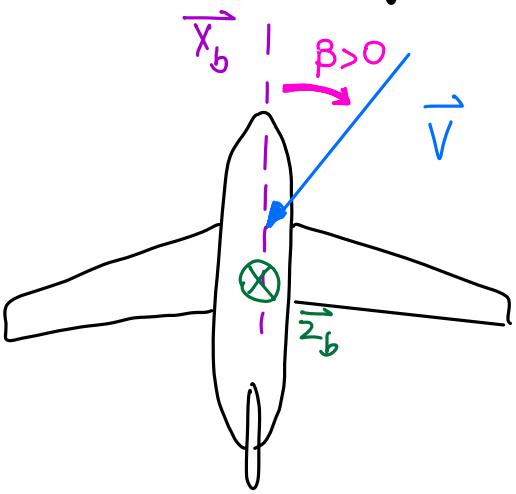
↳ extended to the aircraft in longitudinal motion ($\beta = 0$), that definition becomes the angle between the aerodynamic velocity vector and the fuselage axis



$\alpha > 0$ if $\vec{x}_b$ is ABOVE the $(\vec{x}_a\ \vec{y}_a)$ plane :

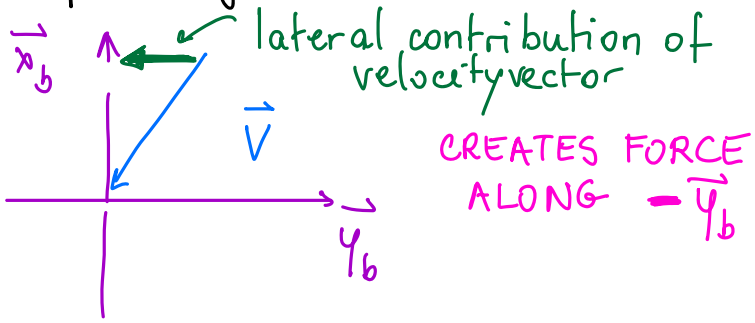
PILOT ABOVE VELOCITY VECTOR

Side-slip angle β (lateral "angle of attack")



defined as the angle between the fuselage axis and the velocity vector if there is no pitch

$\beta > 0$ if air blows on RIGHT side of fuselage

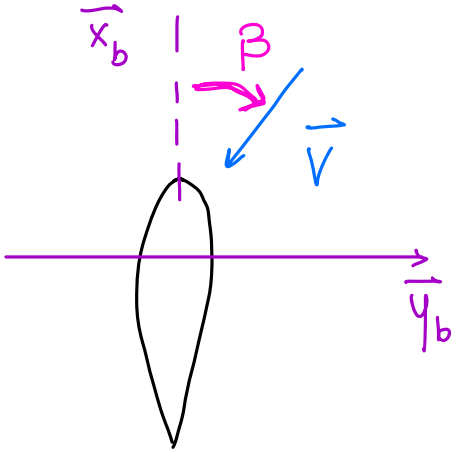


CREATES FORCE ALONG $-\vec{y}_b$

Pilot to the LEFT of the velocity vector,

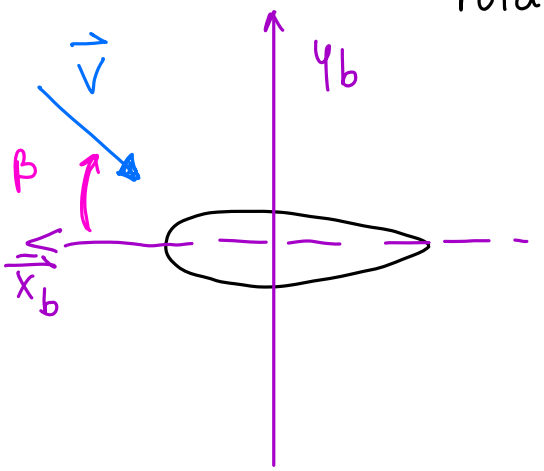
lateral force along $-\vec{y}_b$

Zoom onto vertical stabilizer and compare with main wing

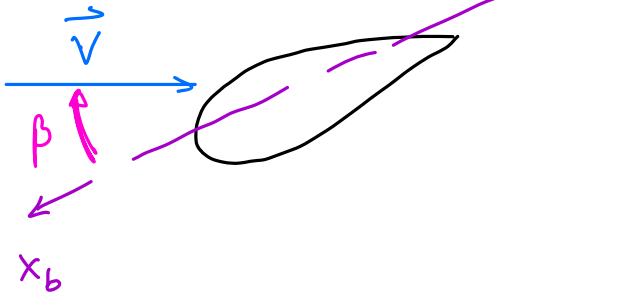


because β is defined from x_b to $\vec{V}$,
a $\beta > 0$ configuration is equivalent to what we would have called a $\alpha < 0$ situation earlier $\Rightarrow$ negative force

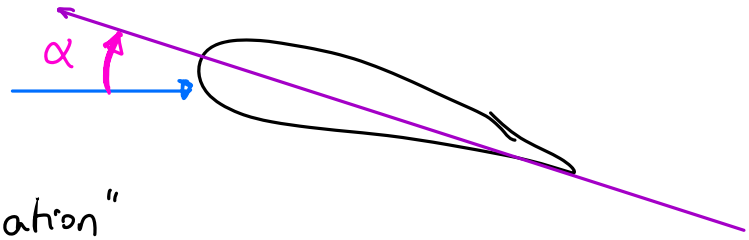
rotate counter clockwise



align velocity vector with horizontal on the page



$\neq$ from typical "airfoil in designed configuration"



The lateral force from side-slip is mostly due to

$$C_{Y_\beta} = -\frac{2\pi}{1+\frac{2}{AR_{VS}}}\frac{S_{VS}}{S}$$

so that the lateral force from VS from a non-zero side slip can be written

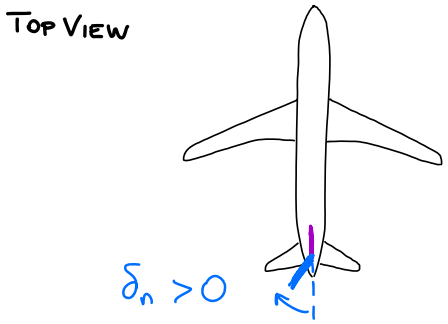
$$Y_{\text{from }\beta} = \frac{1}{2}\rho S V^2 C_{Y_\beta}\beta$$

S: Aircraft reference surface area

$\beta > 0 \rightarrow$ negative lateral force

- Influence of yaw control δ_n

$\delta_n > 0 \Rightarrow$ deflecting rudder to the LEFT



Sign of $C_{Y_{\delta_n}}$

$\delta_n > 0 \rightarrow$ control to the left of symmetry plane

"deploying flaps for VS"-like

$\Downarrow$

lateral force > 0 (along Y_b)

$$\boxed{C_{Y_{\delta_n}} > 0}$$

- Influence of ailerons δ_ℓ : not creating any significant lateral force

$$\boxed{C_{Y_{\delta_\ell}} \approx 0}$$

5.2 Aerodynamic moment coefficients C_ℓ and C_n

5.2.1 Influence of side slip β

Overall lateral force due to side-slip NOT through G

$\Rightarrow$ MOMENTS !

Roll moment

$$\frac{1}{2}\rho S V^2 \ell\, C_{\ell_\beta}\beta$$

Yaw moment

$$\frac{1}{2}\rho S V^2 \ell\, C_{n_\beta}\beta$$

$\Rightarrow$ similar to lift force

$\alpha \Longleftrightarrow \beta$

concept of aerodynamic center for lateral motion

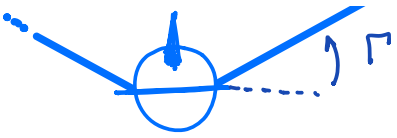
$\Rightarrow$ signs of C_{ℓ_β} and C_{n_β} depend on the aircraft

Roll moment due to side slip - C_{ℓ_β} (Dihedral effect)

Main effect comes from dihedral of wings

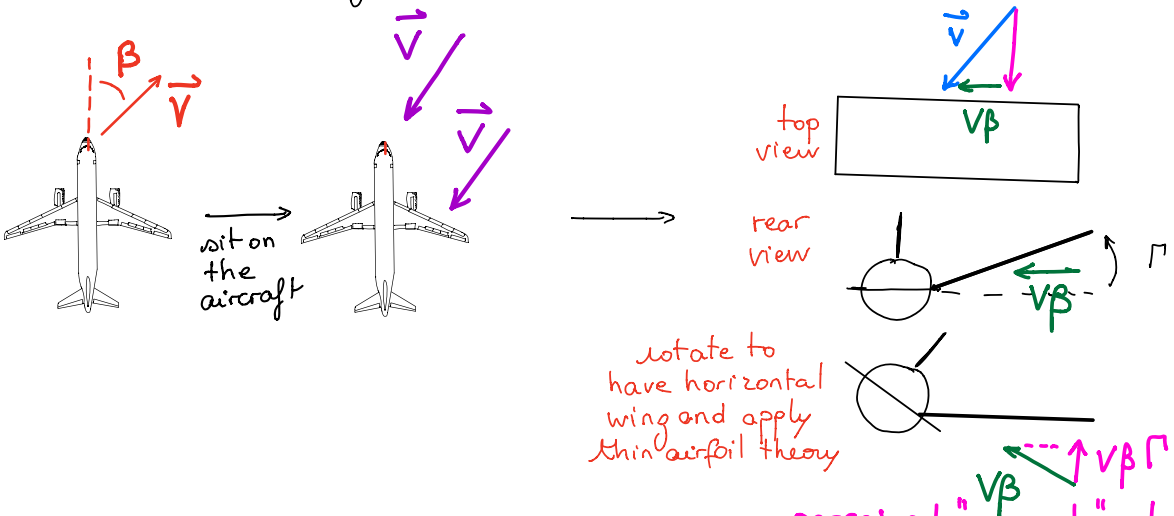
Γ : angle between wings and horizontal

$\Gamma > 0$ if wing tip is higher than wing roots



Physical intuition : side slip will "lift" the right wing, make it roll to the left. $\Rightarrow$ $C_{\ell_\beta} < 0$

Simple model : Switching to a wing perspective :



$\Rightarrow$ perceived "upwash" : larger angle of attack : higher lift on right wing

$\Rightarrow$ roll moment to the left.

Thin airfoil theory :

$$C_{\ell_{\beta_\Gamma}} = -\frac{b}{\ell}\,\frac{1+2\lambda}{1+\lambda}\,\frac{C_{z_\alpha}}{6}\,\Gamma$$

(b: wing span; ℓ: ℓ)

$C_{\ell_{\beta_\Gamma}}$: there are other contributions to C_{ℓ_β}

λ : taper ratio

$$= -\frac{1+3\lambda}{1+\lambda}\,\frac{AR}{8}\,C_{z_\alpha}\,\Gamma$$

$$= -\frac{1+3\lambda}{1+\lambda}\,\frac{\pi}{4}\,\frac{AR^2}{2+AR}\,\Gamma$$

main message : dihedral effect proportional to Γ & depends on AR and taper ratio λ.

Γ is not the only factor impacting C_{ℓ_β}. Among the other main effects are:

- Vertical position of the wings
- Size of vertical empennage
- "Keel" effect (created by any surface above or below the center of gravity G)
- Sweep angle Λ

smaller C_{ℓ_β}, $C_{\ell_\beta} < 0$ → larger C_{ℓ_β}, $C_{\ell_\beta} > 0$

larger dihedral (large Γ)
high wings
backward sweep angle
large vertical stab

small/no/negative dihedral
low wings
no/forward sweep angle
small vertical stab

In practice, from the point of view of lateral flight handling qualities, it is desirable to have: $C_{\ell_\beta} < 0$ but not too much

- closely linked to spiral mode stability (see later)
- aircrafts with low wings tend to have HIGHER Γ than aircrafts with high wings.
- in practice, size of the VS that is typically used to adjust the value of C_{ℓ_β}

Yaw moment due to side slip - C_{n_β}

To look at C_{n_β}, let's decompose the lateral force into two contributions:

Overall force acting on fuselage + Overall force action on vertical stabilizer

smaller contribution acting at a point $A_{fuselage}$ (similar to lift and aerodynamic center)

$A_{fuselage}$ typically in front of G

$\beta > 0 \Rightarrow$ lateral force < 0 in front of G

$\Downarrow$

negative yaw moment

$C_{n_\beta}^{fuselage} < 0$

typically the larger contribution : force applied far from G

$\beta > 0 \Rightarrow$ lateral force < 0

$\Downarrow$

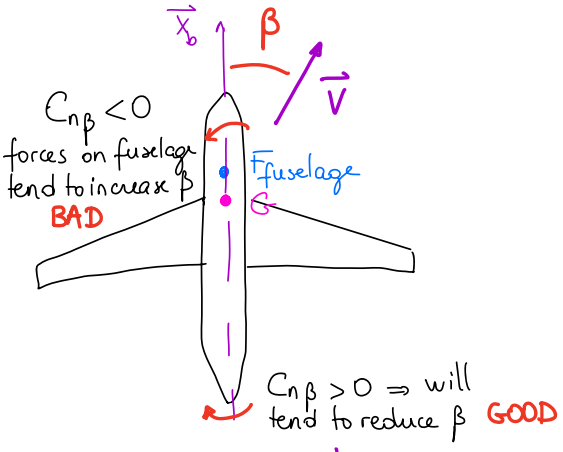
positive yaw moment

$C_{n_\beta}^{VS} > 0$

Which effect is most desirable? $C_{n_\beta} > 0$

because it has a stabilizing effect

"weathervane effect"



any vertical surface IN FRONT OF G will decrease C_{n_β} and have destabilizing effects

$\Rightarrow$ additional constraint on size of V.S.

C_{n_β} can be easily derived from C_{Y_β} using simple relationships between forces and moments:

<0 lateral force, need to get positive moment

DIMENSIONAL $N_\beta = -d\, Y_\beta$ ($Y_\beta \leftarrow C_{Y_\beta}\beta$)

N_β: yaw moment created by side slip

d: distance between G and VS

$Y_\beta = \frac{1}{2}\rho S V^2 C_{Y_\beta}\beta$

$N_\beta = \frac{1}{2}\rho S V^2 \ell\, C_{n_\beta}\beta$

ℓ: mean aero chord

NON-DIMENSIONAL $C_{n_\beta} = -\frac{1}{k_{n_\beta}}\frac{d}{\ell} C_{Y_\beta}$

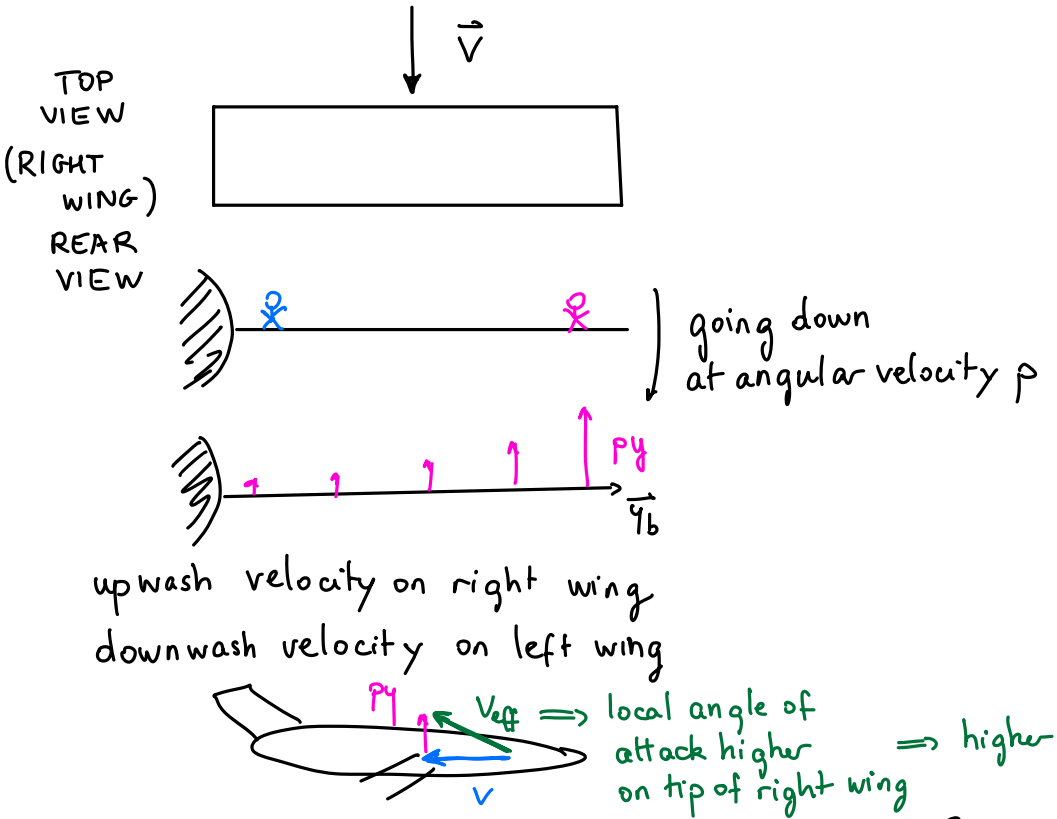
k_{n_β} accounts for destabilization of fuselage

$k_{n_\beta} > 1$: for most aircraft $k_{n_\beta} \approx \frac{d}{\ell} \Rightarrow C_{n_\beta} \approx -C_{Y_\beta}$

5.2.2 Influence of roll velocity p

Roll moment due to roll velocity - C_{ℓ_p}

When roll velocity is positive: $p > 0$, the right wing goes DOWN.



local effective angle of attack on wing going down is higher than the one on the wing going up

$\Downarrow$

roll moment < 0

$\Downarrow$

$C_{\ell_p} < 0$

$V_{eff} \Rightarrow$ local angle of attack higher on tip of right wing $\Rightarrow$ higher lift

- From the lifting line theory, we can express C_{ℓ_p} as:

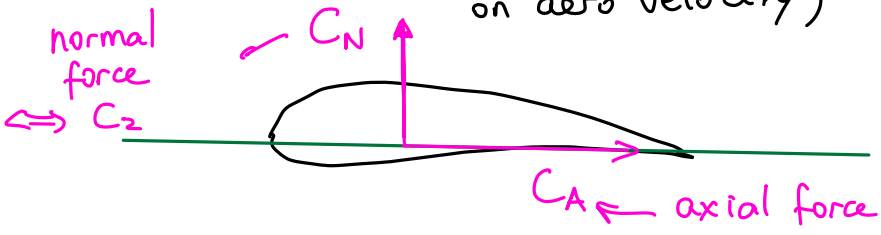
$$C_{\ell_p} = -\frac{\pi}{8}\frac{AR^3}{4+AR}$$

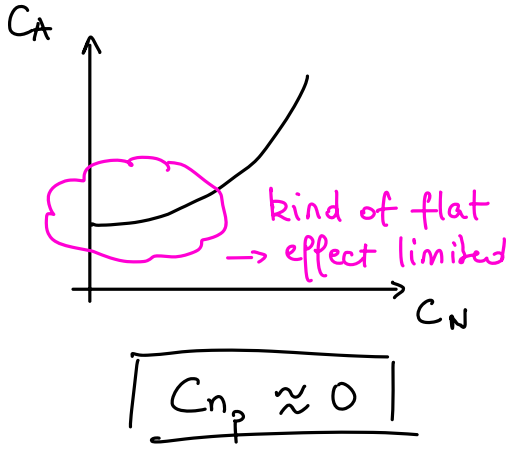
- Other contributions (vertical and horizontal stabilizers) also provide negative contributions to C_{ℓ_p}.

- Roll moment created by roll velocity opposes motion
  ↳ DAMPING effect

Yaw moment due to roll velocity - C_{n_p}

Lilienthal polar : $\equiv$ drag polar but expressed in airfoil axes instead of Ra (based on aero velocity)





$C_{n_p} \approx 0$

5.2.3 Influence of yaw velocity r

Roll moment due to yaw velocity - C_{ℓ_r}

There are two main contributions to C_{ℓ_r}: wing and vertical stabilizer:

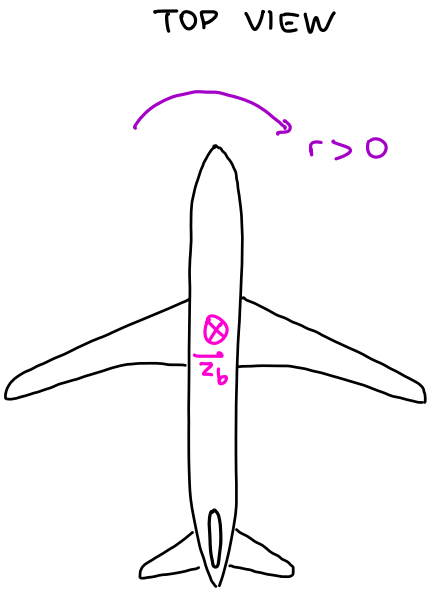
- **Wings:**

$r>0 \rightarrow$ left wing going faster

$\rightarrow$ lift FORCE higher on the left

$\rightarrow$ creates a ROLL moment that brings the RIGHT wing down

$\rightarrow C_{\ell_r}^{WING} > 0$



- **Vertical stabilizer:**

$r>0 \rightarrow$ VS moves to the left

$\rightarrow$ creates a lateral force differential that is >0 (in the direction of y_b)

$\rightarrow$ that force is ABOVE G $\rightarrow$ creates a roll moment that brings the right wing down

$\rightarrow C_{\ell_r}^{VS} > 0 \Rightarrow$ OVERALL $C_{\ell_r} > 0$

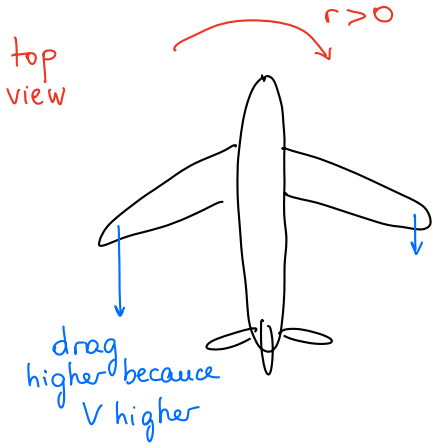
The wings dominate C_{ℓ_r}. The lifting line theory provides an analytical model for it:

$$C_{\ell_r} \approx \frac{1}{16} \frac{1+4\lambda}{1+\lambda} AR^2 C_z$$

Yaw moment due to yaw velocity - C_{n_r}

There are again two main contributions to C_{n_r}: wing and vertical stabilizer:

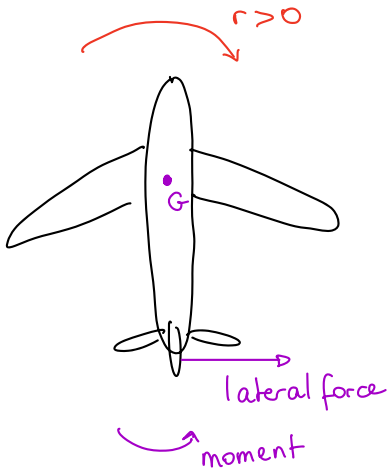
- **Wings:**



left wing goes faster
↓
drags more
↓
create yaw moment that opposes motion (<0)

$$C_{n_r}^{\text{wing}} < 0$$

- **Vertical stabilizer:**



VS "moves to the left"
↓
creates >0 lateral force
↓
creates <0 yaw moment

$$C_{n_r}^{VS} < 0$$

DAMPING ROLE

In this case, the vertical stabilizer dominates C_{n_r}. Using simple force/moment relations, we obtain a simple model for it:

$$C_{n_r} = \left(\frac{d}{\ell}\right)^2 C_{Y_\beta} \left(\frac{l}{k_{n_r}^2}\right)$$

$$k_{n_r}^2 \approx 2$$

5.2.4 Influence of roll control δ_ℓ (ailerons)

2 ailerons on trailing edge of wings

↳ when the right aileron goes down, left one goes up

$\delta_{\ell_{RIGHT}} > 0$ $\delta_{\ell_{LEFT}} < 0$

$$\boxed{\delta_\ell = 0.5\left(\delta_{\ell_{RIGHT}} - \delta_{\ell_{LEFT}}\right)}$$

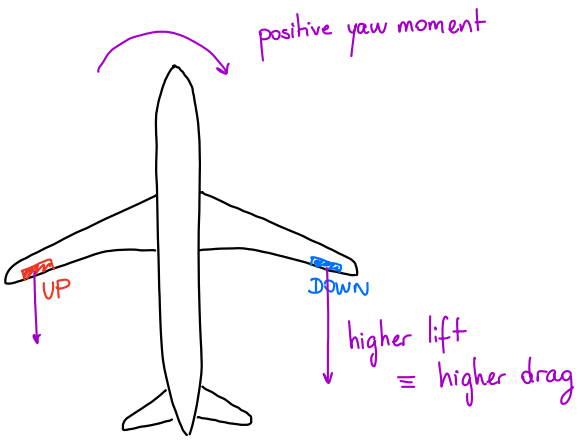
Roll moment due to δ_ℓ - $C_{\ell_{\delta_\ell}}$

$\delta_\ell > 0 \rightarrow$ stick to the left (intuitive to pilot), right aileron down

lift on left wing ↓
lift on right wing ↑
} roll to the left

$$C_{\ell_{\delta_\ell}} < 0$$

Yaw moment due to δ_ℓ - $C_{n_{\delta_\ell}}$



$\delta_\ell > 0$

⇓

{ lift on the left ↓
drag " " ↓

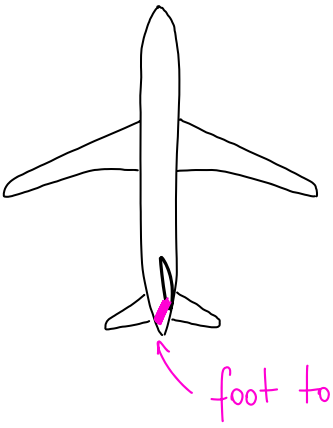
+ { lift on the right ↑
drag " " ↑

⇓

positive yaw moment

$$\boxed{C_{n_{\delta_\ell}} > 0}$$

5.2.5 Influence of yaw control δ_n (rudder)



$\delta_n > 0$ when rudder is on the left of the symmetry plane

Roll moment due to δ_n - $C_{\ell_{\delta_n}}$

$\delta_n > 0 \rightarrow$ lateral force on the VS > 0 and ABOVE center of gravity

↓

roll moment > 0

↓

$$\boxed{C_{\ell_{\delta_n}} > 0}$$

Yaw moment due to δ_n - $C_{n_{\delta_n}}$

$\delta_n > 0 \rightarrow$ lateral force on VS > 0

↓

yaw moment < 0

⇓

$$\boxed{C_{n_{\delta_n}} < 0}$$

6 Summary

Force and Moment Coefficients	Side Slip β	Roll Velocity p	Yaw Velocity r	δ_a	δ_r
Lateral Force C_Y	Lateral lift $C_{Y_\beta} < 0$ "yaw damping" side-slip oscillations	$C_{Y_p} \approx 0$	$C_{Y_r} \approx 0$	$C_{Y_{\delta_a}} \approx 0$	$C_{Y_{\delta_r}} > 0$
Roll Moment C_ℓ	Dihedral effect $C_{\ell_\beta} < 0$ Spiral mode stability	$C_{\ell_p} < 0$ damping of roll mode	$C_{\ell_r} > 0$ coupling yaw/roll	$C_{\ell_{\delta_a}} < 0$	$C_{\ell_{\delta_r}} > 0$ often neglected
Yaw Moment C_n	Route stability effect $C_{n_\beta} > 0$ yaw feedback side-slip oscillations	$C_{n_p} \approx 0$	$C_{n_r} < 0$ yaw damping side-slip oscillations	$C_{n_{\delta_a}} > 0$ often neglected	$C_{n_{\delta_r}} < 0$